

Unit 6: Limits Involving Infinity; Asymptotes of Graphs

1. The idea, in plain words

This unit asks two closely related questions:

1. “What happens to $f(x)$ as x grows forever, in either direction?” — this is a limit at infinity, and it tells us about horizontal asymptotes: flat lines the graph flattens out toward (but doesn’t necessarily touch) way out on the left or right.
2. “What happens to $f(x)$ as x approaches some specific number where the function blows up?” — this connects straight back to the infinite discontinuities from Unit 5, and tells us about vertical asymptotes: vertical lines the graph shoots up or down alongside, without ever crossing.

Picture it: imagine a graph that, way off to the right, gets flatter and flatter, hugging a horizontal line more and more closely — that flat line is the horizontal asymptote. Now imagine a totally different spot on the same graph where, as you approach one particular x -value, the curve rockets straight up (or down) like a rocket ship — that’s a vertical asymptote.

The key trick for limits at infinity with fractions of polynomials: you can’t just “plug in infinity” — infinity isn’t a number. Instead, we divide every single term (top and bottom) by the highest power of x found in the denominator. Every term that ends up with an x in the bottom (like $\frac{1}{x}$ or $\frac{5}{x^2}$) shrinks to 0 as $x \rightarrow \infty$, since dividing by a bigger and bigger number gives you less and less. What’s left tells you the answer instantly.

2. Toolbox

Limits of simple power terms:

$$\lim_{x \rightarrow \infty} \frac{c}{x^n} = 0 \quad \text{for any constant } c \text{ and any } n > 0$$

The three-case rule for rational functions (fraction of polynomial over polynomial) as $x \rightarrow \infty$ or $x \rightarrow -\infty$ — compare the degree (highest power of x) on top versus bottom:

- Top degree < bottom degree: the limit is 0. Horizontal asymptote: $y = 0$.
- Top degree = bottom degree: the limit is the ratio of the leading coefficients (the numbers in front of the highest powers). Horizontal asymptote: $y = \frac{\text{leading coeff. of top}}{\text{leading coeff. of bottom}}$.
- Top degree > bottom degree: the limit is $+\infty$ or $-\infty$ (the fraction grows without bound). There is no horizontal asymptote in this case.

How to actually compute it: divide every term in the numerator and denominator by x^n , where n is the highest power appearing in the denominator. Then apply $\lim_{x \rightarrow \infty} \frac{c}{x^k} = 0$ to every leftover fraction term.

Vertical asymptotes: for a rational function $\frac{P(x)}{Q(x)}$, a vertical asymptote occurs at any $x = a$ where $Q(a) = 0$ and $P(a) \neq 0$ (the factor causing the zero does not cancel out). This is exactly the infinite-discontinuity idea from Unit 5.

The cancellation trap (important!): always factor the top and bottom fully before deciding on vertical asymptotes. If a factor cancels between the numerator and denominator, that x -value is only a removable hole, not a vertical asymptote — the graph doesn’t blow up there at all.

3. Common mistakes

- Trying to “plug in infinity.” Infinity is not a number you can substitute — always divide by the highest power in the denominator instead.
- Forgetting to factor before finding vertical asymptotes. If you skip factoring, you might mistakenly call a removable hole a vertical asymptote (or vice versa).
- Mixing up which direction the graph shoots — $+\infty$ or $-\infty$. Check the sign of the expression just barely to the left and just barely to the right of the asymptote separately; they can be different.
- Assuming top degree = bottom degree always gives a “nice” horizontal asymptote of 1. It’s the ratio of leading coefficients, not automatically 1 — don’t forget to actually read off the coefficients.
- Forgetting that a graph can cross its horizontal asymptote (just not near $\pm\infty$) — unlike a vertical asymptote, which the graph never touches at all.

4. Problem Set

Warm-up

1. $\lim_{x \rightarrow \infty} \frac{3x+5}{2x-1}$
2. $\lim_{x \rightarrow \infty} \frac{2x+1}{x^2+3}$
3. $\lim_{x \rightarrow \infty} \frac{x^3+2}{5x^2-1}$
4. $\lim_{x \rightarrow -\infty} \frac{4x^2-3}{2x^2+7}$
5. $\lim_{x \rightarrow \infty} \frac{5}{x^3}$
6. $\lim_{x \rightarrow \infty} \frac{7x^2-2x+1}{3x^2+4x-5}$

Standard

7. $\lim_{x \rightarrow \infty} \frac{5x^2+8x-3}{3x^2+2}$
8. Find the horizontal asymptote of $y = \frac{2x-7}{x+4}$.
9. Find the horizontal asymptote of $y = \frac{3x^2+1}{x^2-5x+6}$.
10. Find the horizontal asymptote of $y = \frac{x+2}{x^2-9}$.
11. Find the vertical asymptote(s) of $y = \frac{x+1}{x^2-4}$.
12. Find the vertical asymptote(s) of $y = \frac{2x}{x^2+x-6}$.

Challenge

13. Find the horizontal and vertical asymptotes of $y = \frac{(x+3)(x+1)}{(x+2)(x-1)}$.

14. Find the horizontal and vertical asymptotes of $y = \frac{(x-2)(x+5)}{(x-2)(x+3)}$. (Watch carefully for a factor that cancels!)
15. Find the horizontal and vertical asymptotes of $y = \frac{x^2-1}{x^2-4x+3}$. (Factor everything first.)
16. Find the horizontal and vertical asymptotes of $y = \frac{3x^2-12}{x^2-x-2}$.
17. For $y = \frac{1}{x-2}$: find the horizontal asymptote, and separately determine whether the graph shoots toward $+\infty$ or $-\infty$ as $x \rightarrow 2^-$ and as $x \rightarrow 2^+$.

Applied

18. A company's average cost per unit (in dollars) when producing x units is $C(x) = \frac{500+3x}{x}$. Find the horizontal asymptote as $x \rightarrow \infty$, and explain what it means about the long-run average cost per unit as production grows very large.
19. A drug's concentration in the bloodstream (mg/L) t hours after being administered is modeled by $C(t) = \frac{50t}{t^2+25}$. Find the horizontal asymptote as $t \rightarrow \infty$, and explain what it means about the drug concentration over a very long time.
20. A city's population (in thousands) t years from now is modeled by $P(t) = \frac{200t+500}{t+2}$. Find the horizontal asymptote as $t \rightarrow \infty$, and explain what it suggests about the population in the very long run.
21. A factory's production rate (units per hour) as a function of available resources x is modeled by $R(x) = \frac{10x}{x+5}$. Find the horizontal asymptote as $x \rightarrow \infty$, and explain what it means about the maximum achievable production rate.